

No-Smallness Similarity for Finite-Dimensional Admissible Modules

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Source Conjecture

Baskakov, Krishtal, and Uskova study a closed operator $A : D(A) \subseteq \mathcal{X} \rightarrow \mathcal{X}$ and admissible perturbation modules in the method of similar operators [1]. In Remark 5.16 on PDF page 23 of arXiv:2404.01227 they conjecture:

Remark 5.16 *In the setting of this section, all of the examples for Theorem 4.8 that we know can be reduced to the case of hypercausal operators. In view of the above result, we conjecture that for A -admissible modules $\mathcal{M}$ such that $J = \mathcal{T}(\varphi_a) = 0$, we have that all operators $A - B$, $B \in \mathcal{M}$, are similar to A without any additional assumptions on B .*

In the harmonious setting of their Section 5, the transforms are

$$JX = \mathcal{T}_0(\varphi_a)X, \quad \Gamma X = \mathcal{T}_0(\psi_a)X,$$

and both map $\mathcal{M}$ into itself. The conjecture asks whether $J = 0$ alone removes every size restriction on B .

Result and Scope

The theorem below proves the conjecture for every finite-dimensional admissible module. It is stated for the slightly more abstract data actually used in the proof. The assumption $\Gamma(\mathcal{M}) \subseteq \mathcal{M}$ holds automatically for the harmonious transforms above, but is not part of the most general abstract definition of an admissible triple.

Proof Intuition

When $J = 0$, the defining commutator identity says that Γ is a right inverse for the commutator with A . On a finite-dimensional Γ -invariant module, injectivity makes Γ bijective. Admissibility then unexpectedly gives much more structure: the perturbation space is an associative algebra and $\Gamma^{-1} = [A, \cdot]$ is a nonsingular derivation of that algebra.

A finite-dimensional algebra with a nonsingular derivation must be nilpotent. Indeed, the derivation preserves the Jacobson radical; if the semisimple quotient were nonzero, its identity would lie in the kernel of the induced derivation. Nilpotence is precisely what is needed to replace the paper's contraction argument by finite algebra. The Friedrichs fixed point series terminates, and the similarity transform has a finite polynomial inverse. Thus no norm bound on B is needed.

Theorem 1 (Finite-dimensional case of Remark 5.16). *Let $A : D(A) \subseteq \mathcal{X} \rightarrow \mathcal{X}$ be a closed operator on a complex Banach space, and let $(\mathcal{M}, 0, \Gamma)$ be an admissible triple for A in the sense of Definition 4.2 of [1]. Assume in addition that*

$$\dim \mathcal{M} < \infty \quad \text{and} \quad \Gamma(\mathcal{M}) \subseteq \mathcal{M}.$$

Then, for every $B \in \mathcal{M}$, the operator $A - B$, with domain $D(A)$, is similar to A . No norm restriction on B is required.

More explicitly, there is an integer q with $\mathcal{M}^q = \{0\}$. For

$$L_B X = B \Gamma X \quad \text{and} \quad X_* = \sum_{n=0}^{q-1} L_B^n B,$$

the sum is finite, $U = I + \Gamma X_$ is boundedly invertible, $U(D(A)) = D(A)$, and*

$$(A - B)U = UA \quad \text{on } D(A).$$

Lemma 1 (Nonsingular derivations force nilpotence). *Let $\mathcal{A}$ be a finite-dimensional associative algebra over $\mathbb{C}$, not necessarily unital. If $D : \mathcal{A} \rightarrow \mathcal{A}$ is an invertible derivation, then $\mathcal{A}$ is nilpotent.*

Proof. Let $R = \text{Rad}(\mathcal{A})$. For each $t \in \mathbb{C}$, the exponential e^{tD} is an algebra automorphism: the derivation rule, applied to the power series, gives

$$e^{tD}(xy) = e^{tD}(x)e^{tD}(y).$$

The Jacobson radical is invariant under every algebra automorphism, hence $e^{tD}R = R$. Differentiating at $t = 0$ gives $D(R) \subseteq R$. Because D is injective and R is finite-dimensional, in fact $D(R) = R$. Therefore D induces an invertible derivation $\overline{D}$ on $\mathcal{A}/R$.

If $\mathcal{A}/R \neq 0$, this finite-dimensional semisimple algebra has an identity 1. Every derivation of a unital algebra kills its identity, since

$$\overline{D}(1) = \overline{D}(1 \cdot 1) = 2\overline{D}(1).$$

This contradicts invertibility of $\overline{D}$. Hence $\mathcal{A} = R$. A finite-dimensional Jacobson-radical algebra is nilpotent, so $\mathcal{A}^q = \{0\}$ for some q . $\square$

Proof of Theorem 1. Admissibility with $J = 0$ gives

$$(\Gamma X)D(A) \subseteq D(A), \quad A\Gamma X - \Gamma X A = X \quad (X \in \mathcal{M}). \quad (1)$$

In particular, Γ is injective. Since it maps the finite-dimensional space $\mathcal{M}$ into itself, it is bijective on $\mathcal{M}$. Consequently every $Y \in \mathcal{M}$ equals ΓX for some $X \in \mathcal{M}$. Thus every element of $\mathcal{M}$ is bounded and maps $D(A)$ into itself.

The space $\mathcal{M}$ is an associative algebra under operator multiplication. Indeed, given $Y, Z \in \mathcal{M}$, choose $W \in \mathcal{M}$ with $Z = \Gamma W$. Condition 4 of admissibility gives

$$YZ = Y\Gamma W \in \mathcal{M}.$$

Define $D = \Gamma^{-1} : \mathcal{M} \rightarrow \mathcal{M}$. Equation (1) identifies

$$D(Y) = AY - YA \quad (Y \in \mathcal{M}).$$

Because each Y preserves $D(A)$, the usual commutator calculation on $D(A)$ is legitimate and gives

$$D(YZ) = D(Y)Z + YD(Z).$$

Thus D is an invertible derivation of $\mathcal{M}$. Lemma 1 yields $\mathcal{M}^q = \{0\}$ for some q .

Each algebra power $\mathcal{M}^k$ is invariant under D , by the derivation rule. Since D is injective and $\mathcal{M}^k$ is finite-dimensional, $D(\mathcal{M}^k) = \mathcal{M}^k$; hence

$$\Gamma(\mathcal{M}^k) = \mathcal{M}^k. \quad (2)$$

For fixed $B \in \mathcal{M}$, let $L_B X = B\Gamma X$. From (2),

$$L_B(\mathcal{M}^k) \subseteq \mathcal{M}^{k+1}.$$

It follows that $L_B^q = 0$. Therefore

$$X_* = (I - L_B)^{-1}B = \sum_{n=0}^{q-1} L_B^n B$$

is a finite sum and satisfies the Friedrichs equation

$$X_* = B\Gamma X_* + B. \quad (3)$$

Put $N = \Gamma X_* \in \mathcal{M}$. Since $\mathcal{M}^q = 0$, one has $N^q = 0$, and therefore

$$U = I + N, \quad U^{-1} = I - N + N^2 - \dots + (-1)^{q-1} N^{q-1}.$$

Both are bounded. Equation (1) shows that $N(D(A)) \subseteq D(A)$; hence every power of N preserves $D(A)$. Thus both U and U^{-1} preserve $D(A)$, and consequently $U(D(A)) = D(A)$.

Finally, using (1) and (3), for $x \in D(A)$,

$$\begin{aligned} (A - B)U &= A + A\Gamma X_* - B - B\Gamma X_* \\ &= A + \Gamma X_* A + X_* - B - B\Gamma X_* \\ &= (I + \Gamma X_*)A = UA. \end{aligned}$$

This is similarity in the source paper's sense. □

Verification Notes

The argument was checked against all five conditions in Definition 4.2 and against the harmonious transform definition in equation (5.1) of the source. The two points not available for an arbitrary abstract admissible triple are made explicit in the theorem: finite dimensionality and $\Gamma(\mathcal{M}) \subseteq \mathcal{M}$.

The proof avoids convergence estimates entirely. The fixed-point equation is solved by an exact finite Neumann sum because L_B raises algebra powers, and invertibility of the intertwiner is an exact finite polynomial calculation. The domain condition is checked in both directions through the polynomial formulas for U and U^{-1} .

No computational experiment is used as evidence. As elementary stress tests, strictly triangular matrix modules reproduce the expected terminating series, while a module containing a directed matrix cycle cannot satisfy the hypotheses: the resulting finite-dimensional algebra would not be nilpotent and hence could not carry the nonsingular commutator derivation above.

Limitations and Novelty Check

This is a complete proof only for finite-dimensional admissible modules. It does not establish surjectivity of Γ , nilpotence, or convergence of the Friedrichs series in a general infinite-dimensional module, so the full conjecture of Remark 5.16 remains open.

Before promotion, the run's registry, solution, attempt, and proof-gap indexes were searched for arXiv:2404.01227, the paper title, and the core similarity terms; no duplicate result was found. A bounded web/arXiv search on 2026-07-21 used the exact paper title, the phrase "A-admissible modules," the arXiv id, and the stated similarity conclusion. It found the 2024 source paper but no later arXiv paper explicitly resolving Remark 5.16 or this finite-dimensional subcase. This is a bounded novelty check, not a claim of exhaustive literature coverage.

Human Review Recommendation

Review as a strong partial result: it proves the conjectured no-smallness conclusion for every finite-dimensional harmonious admissible module. The highest-value checks are the passage from Γ -bijectivity to algebra closure, invariance of the Jacobson radical under D , preservation of $\mathcal{M}^k$ by $\Gamma = D^{-1}$, and the two-sided domain invariance of the polynomial intertwiner. If these checks pass, the packet is suitable for promotion as a finite-dimensional theorem; it should not be labeled a full solution of the unrestricted conjecture.

References

- [1] A. G. Baskakov, I. A. Krishtal, and N. B. Uskova, *Method of similar operators in harmonious Banach spaces*, arXiv:2404.01227 (2024).